Image and Video Generations

Lecture 5: Zero-Shot Applications

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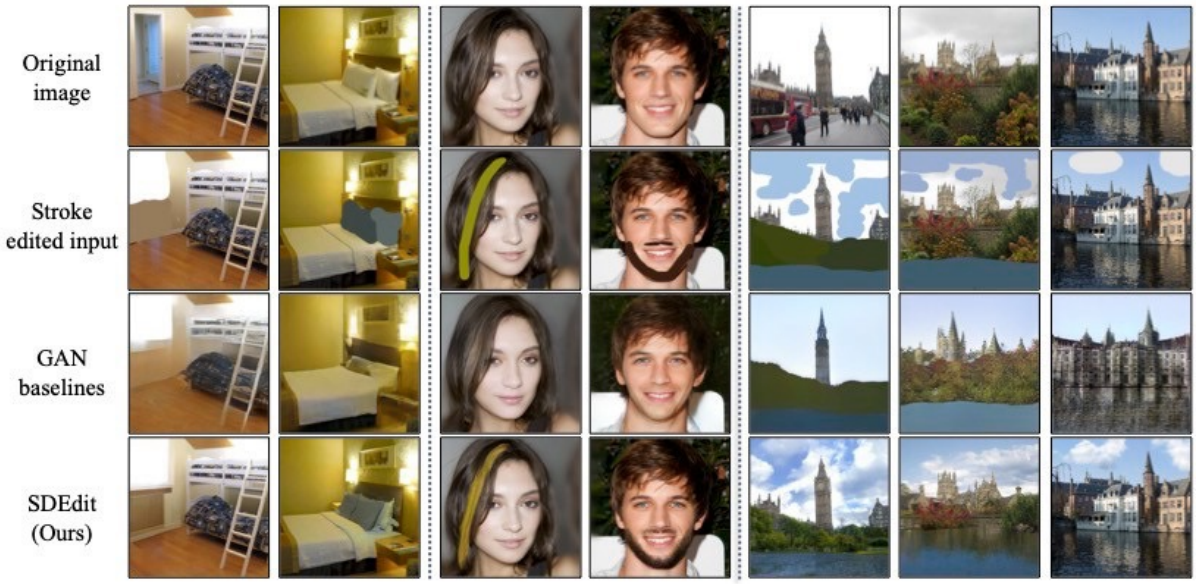
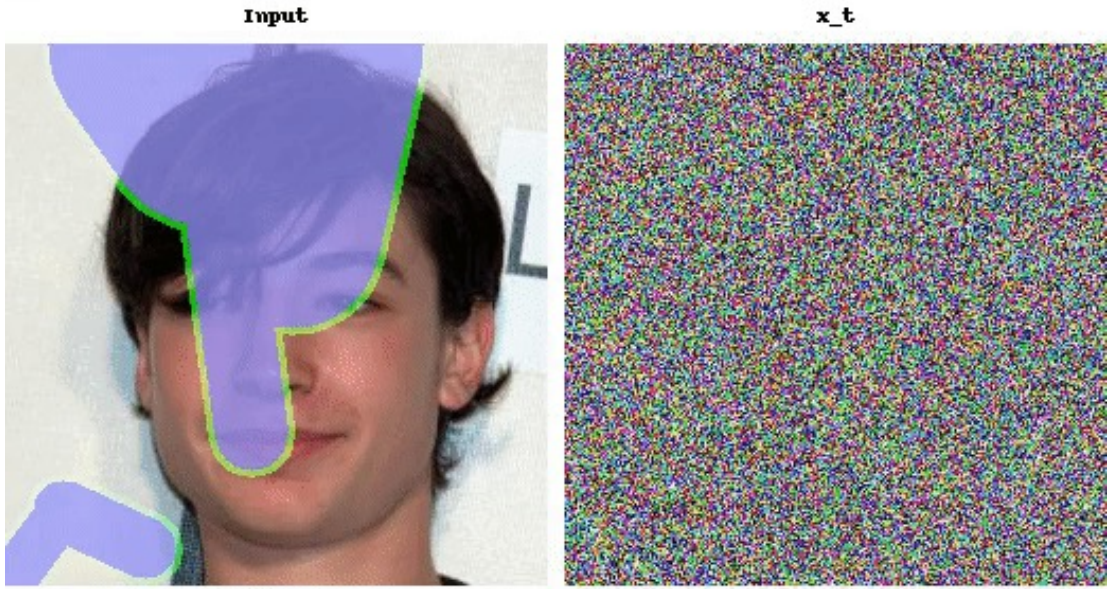
Zero-Shot Applications

Zero-Shot Adaptations

How to edit and inpaint images using a pretrained image diffusion model even without the need for fine-tuning?



Meng et al., SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations, ICLR 2022.

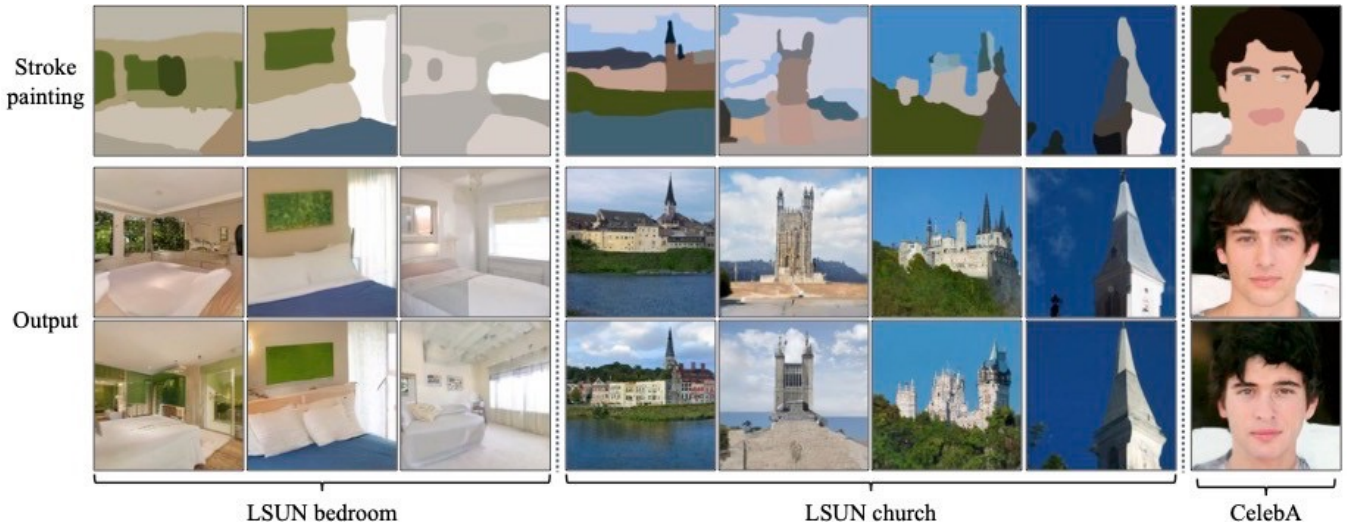


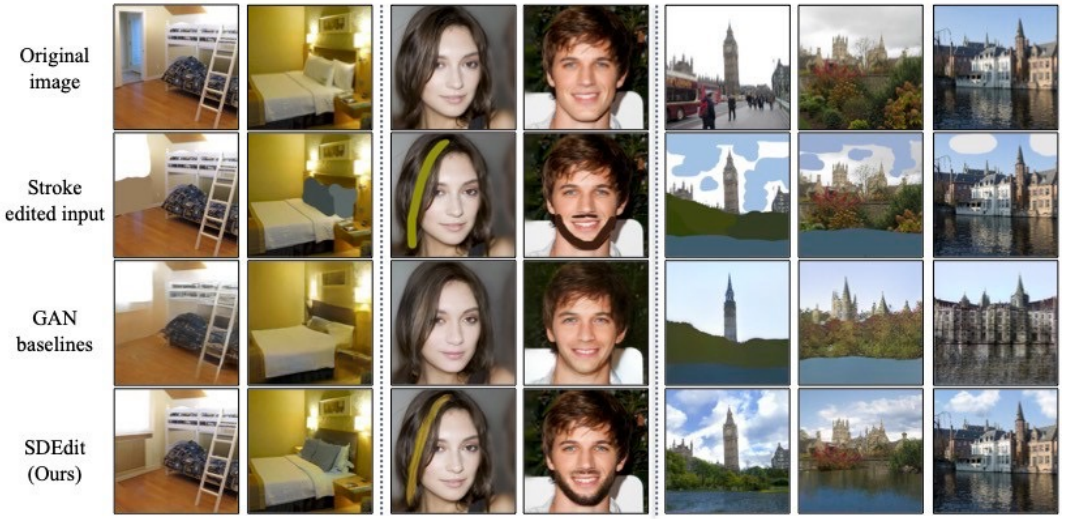
Lugmayr et al., RePaint: Inpainting using Denoising Diffusion Probabilistic Models, CVPR 2022.

SDEdit

Image generation/editing through user interaction

- Image generation from sketches
- Image editing from scribbles

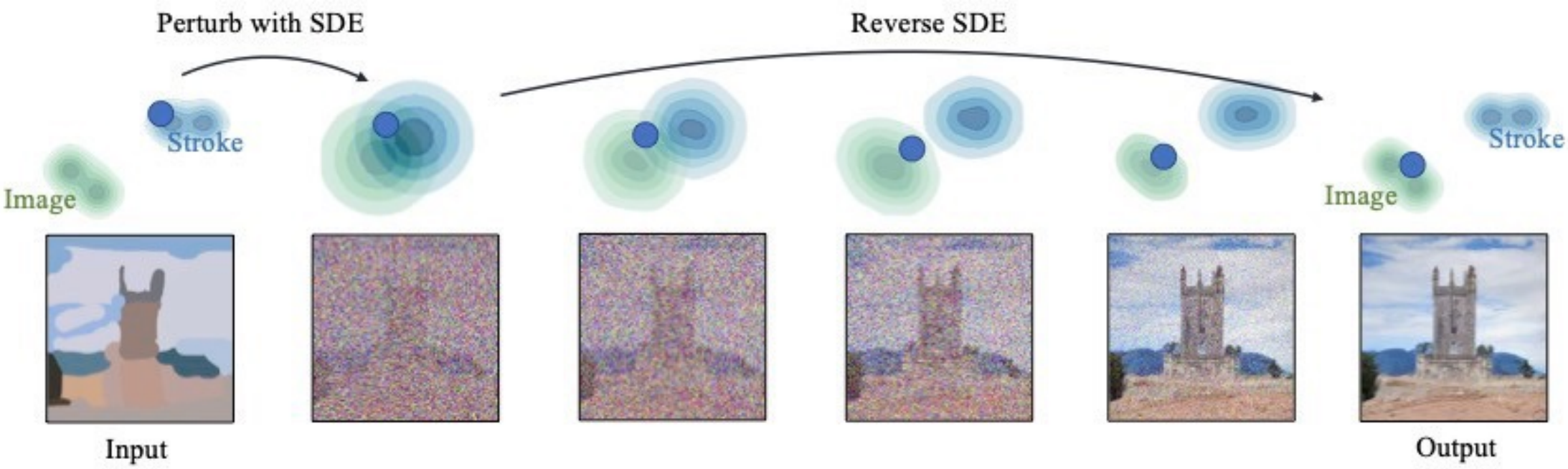




Meng, Chenlin, et al. "SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations." *ICLR.* 2022.

SDEdit

Perform the forward process for a bit and then reverse the process.



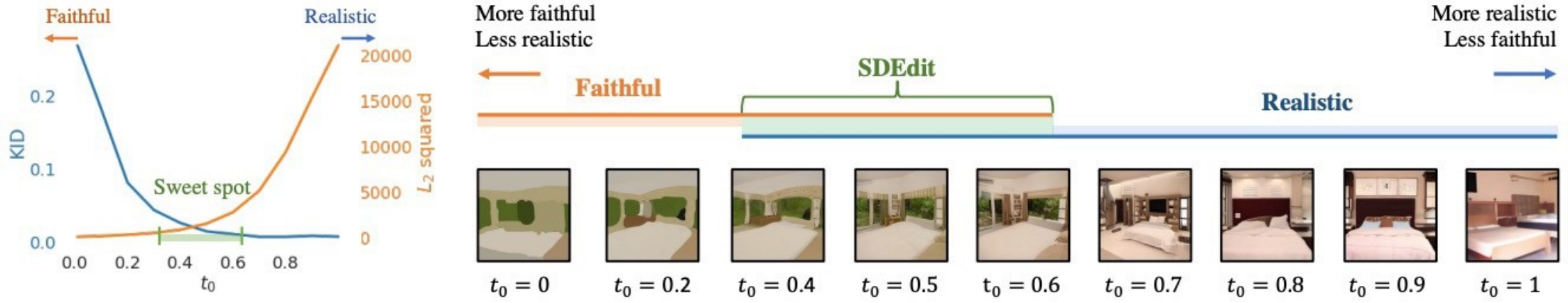
用 Noise 把 input 稍微淹沒，再讓 diffusion 去生成自然影像的細節

Meng, Chenlin, et al. "SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations." *ICLR.* 2022.

SDEdit

Realism vs. faithfulness

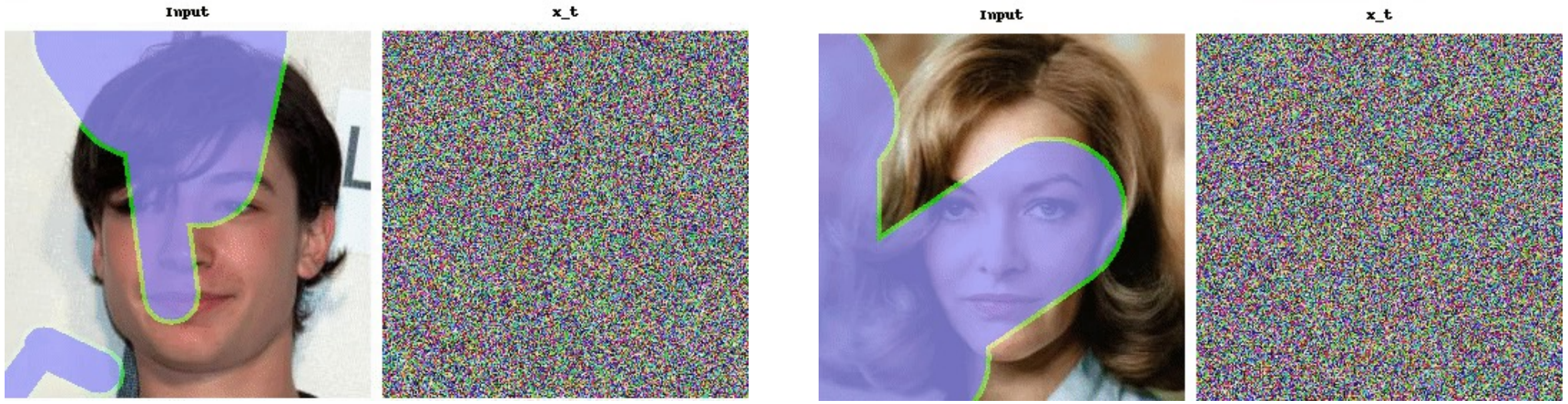
As you perform the forward process with a longer timestep, the output image becomes more realistic but less faithful (deviating from the given condition).



Meng, Chenlin, et al. "SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations." *ICLR.* 2022.

RePaint

- Filling regions in an image using a pretrained image diffusion model.
- Assume that the missing foreground is filled while the background is fixed.



RePaint

Combine denoised foreground images (to be filled) and noisy background (to be fixed) images at each iteration of the reverse process.

想保留的部分

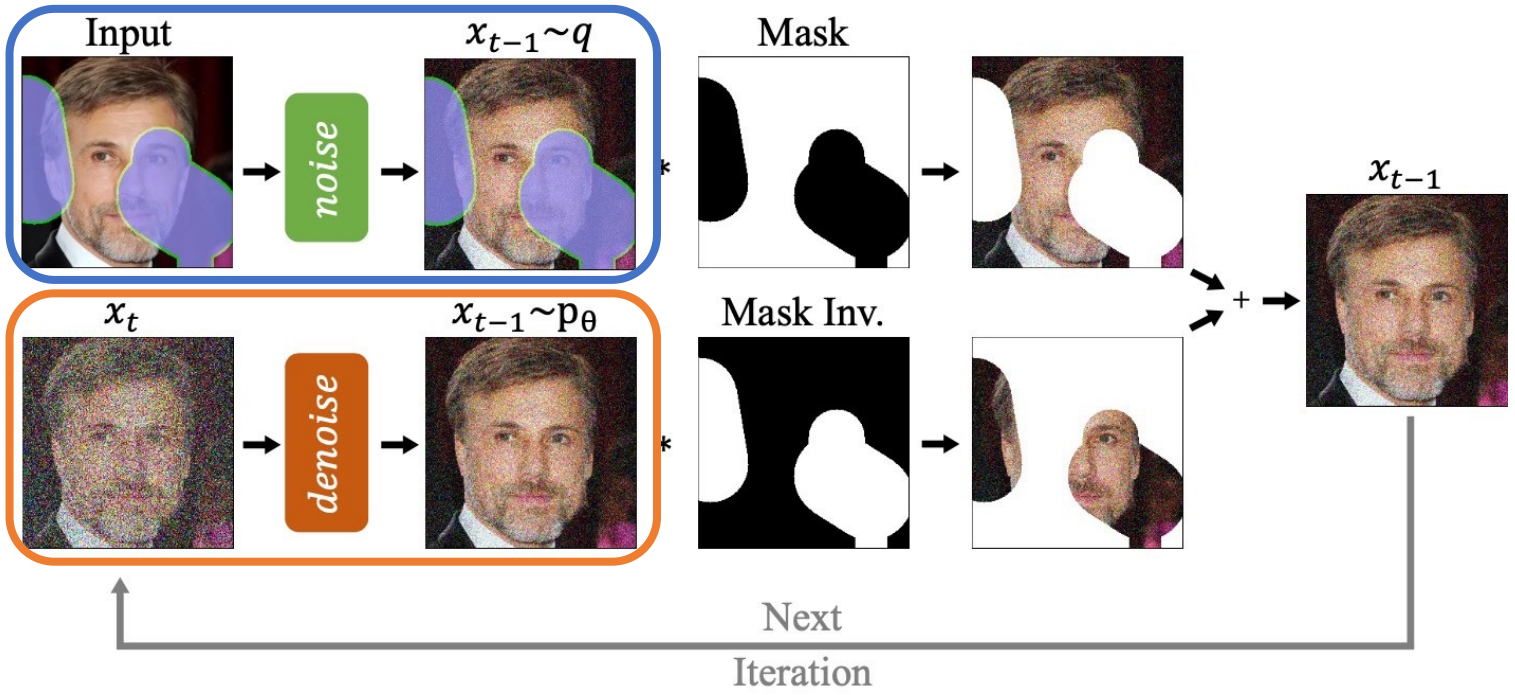
Background

從最乾淨的 x_0 出發

想生成的部分

Foreground

從最 noisy 的 x_T 出發



RePaint

1. Starting from $\boldsymbol{x}_T$, at each timestep t, denoise $\boldsymbol{x}_t$ one step (reverse process), resulting $\boldsymbol{x}_{t-1}^{f}$.

想保留的部分

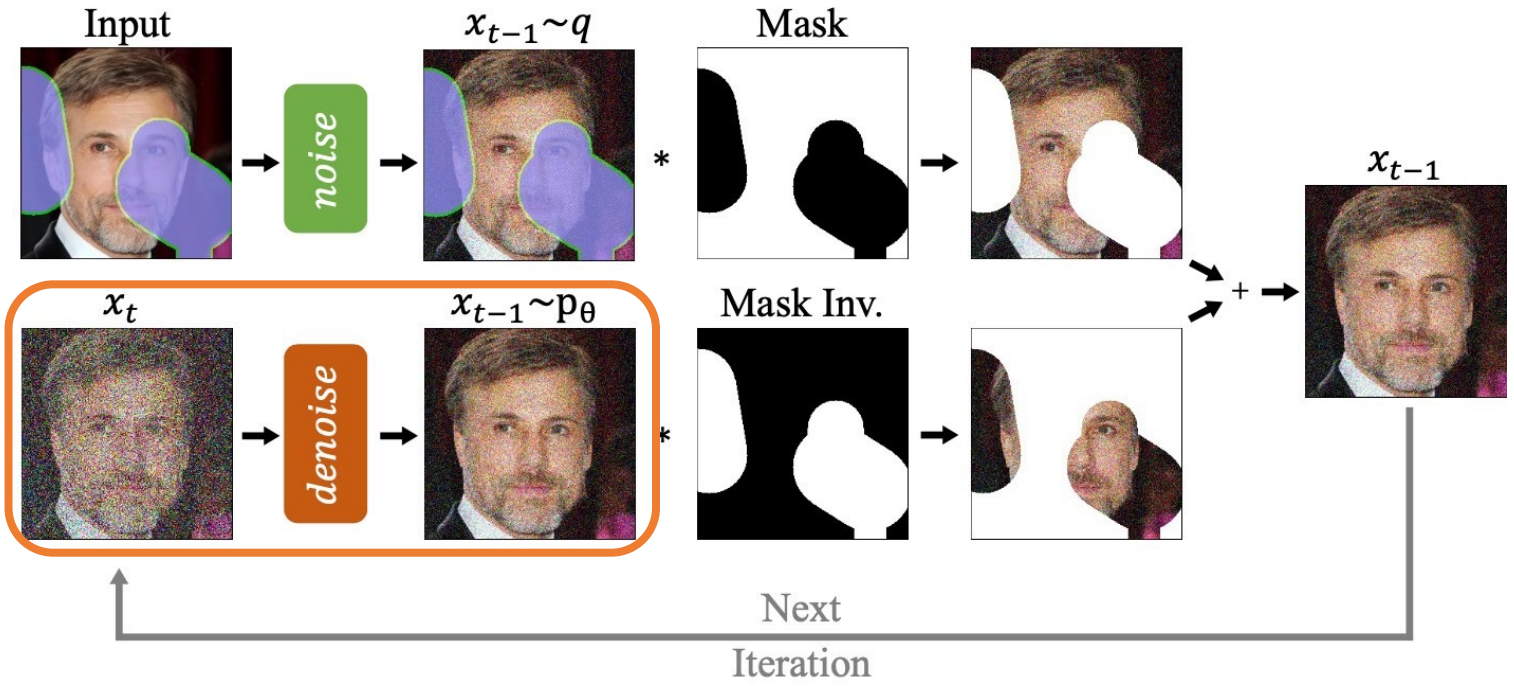
Background

從最乾淨的 $\boldsymbol{x}_0$ 出發

想生成的部分

Foreground

從最 noisy 的 $\boldsymbol{x}_T$ 出發



RePaint

2. Perturb the input background image $\boldsymbol{x}_0^b$ (forward process) with a noise scale of the timestep $t-1$, resulting $\boldsymbol{x}_{t-1}^b$.

想保留的部分

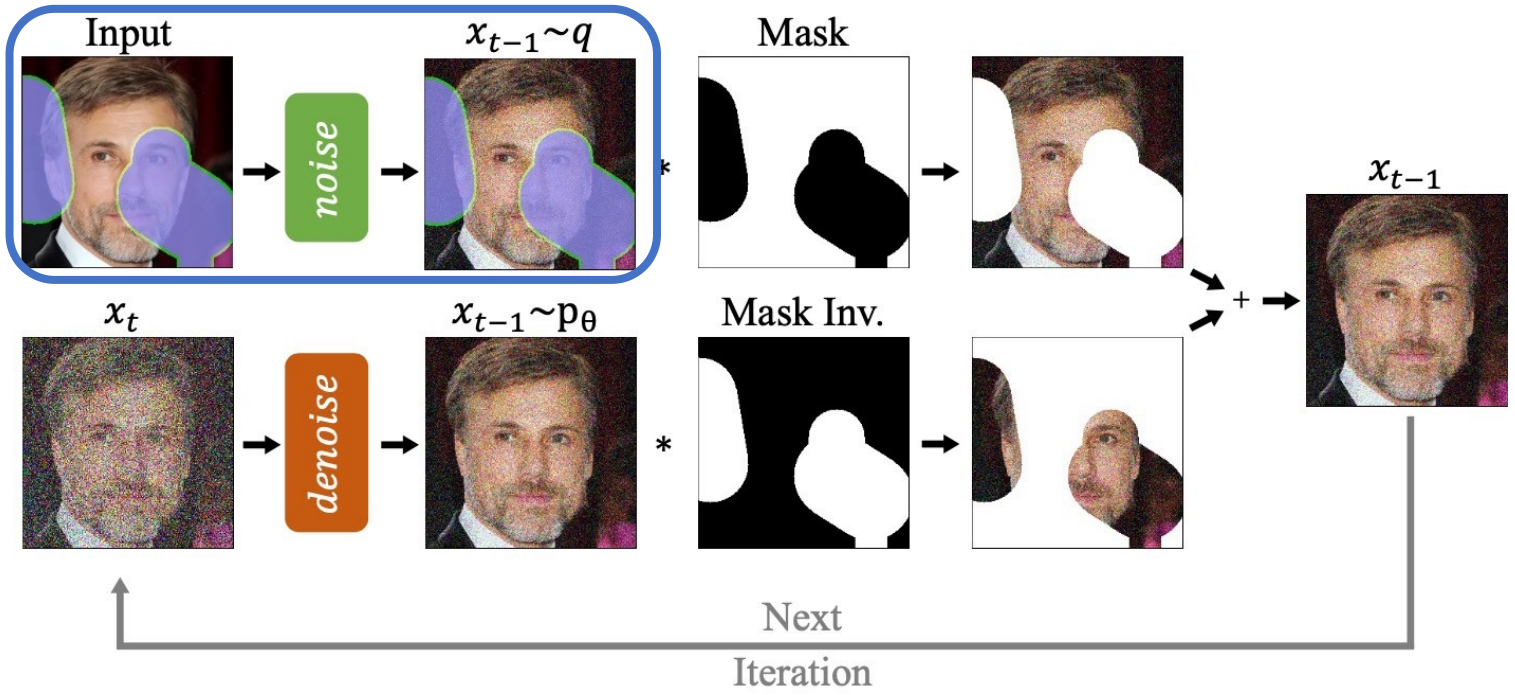
Background

從最乾淨的 $\boldsymbol{x}_0$ 出發

想生成的部分

Foreground

從最 noisy 的 $\boldsymbol{x}_T$ 出發



RePaint

3. Combine $\boldsymbol{x}_{t-1}^{b}$ and $\boldsymbol{x}_{t-1}^{f}$ (where M is the background mask):

$$\boldsymbol{x}_{t-1} = M \odot \boldsymbol{x}_{t-1}^{b} + (1 - M) \odot \boldsymbol{x}_{t-1}^{f}$$

想保留的部分

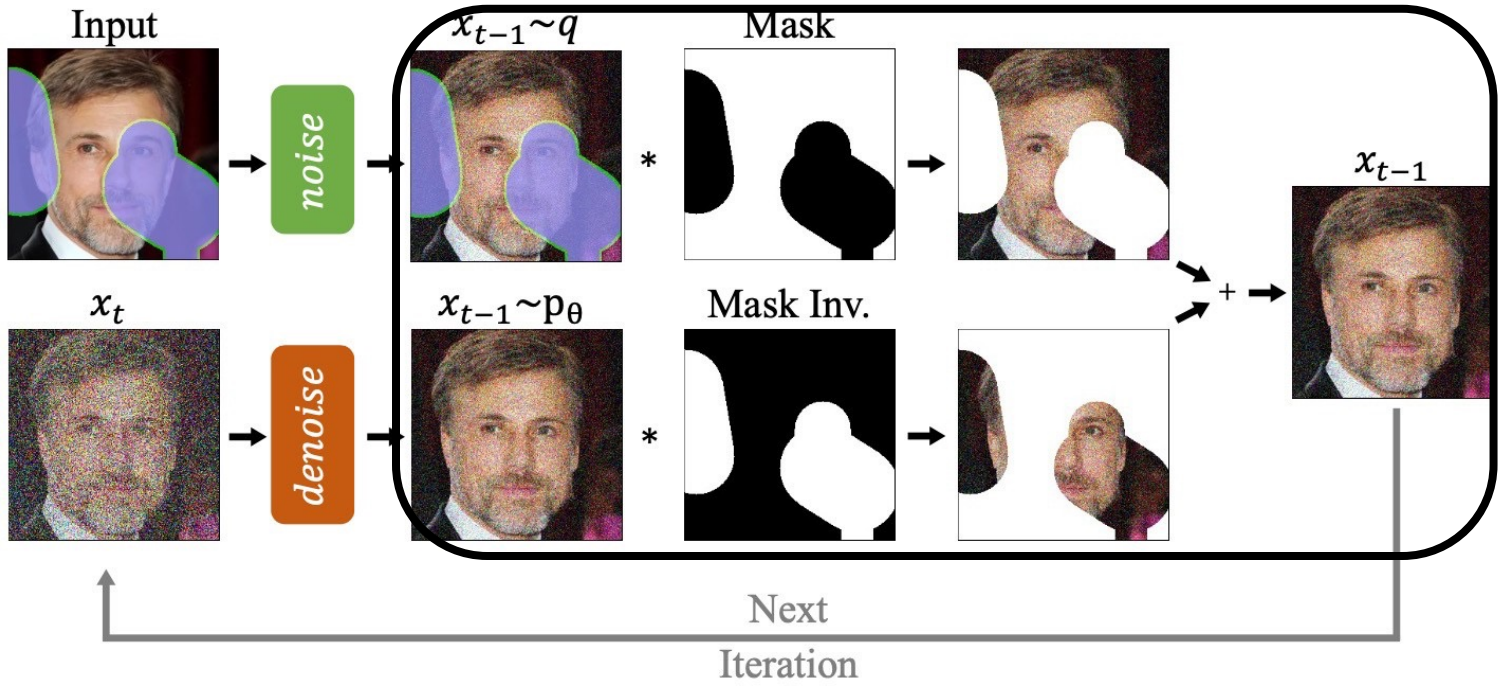
Background

從最乾淨的 $\boldsymbol{x}_0$ 出發

想生成的部分

Foreground

從最 noisy 的 $\boldsymbol{x}_T$ 出發



RePaint

4. Repeat this process for $t = T, \ldots, 1$.

想保留的部分

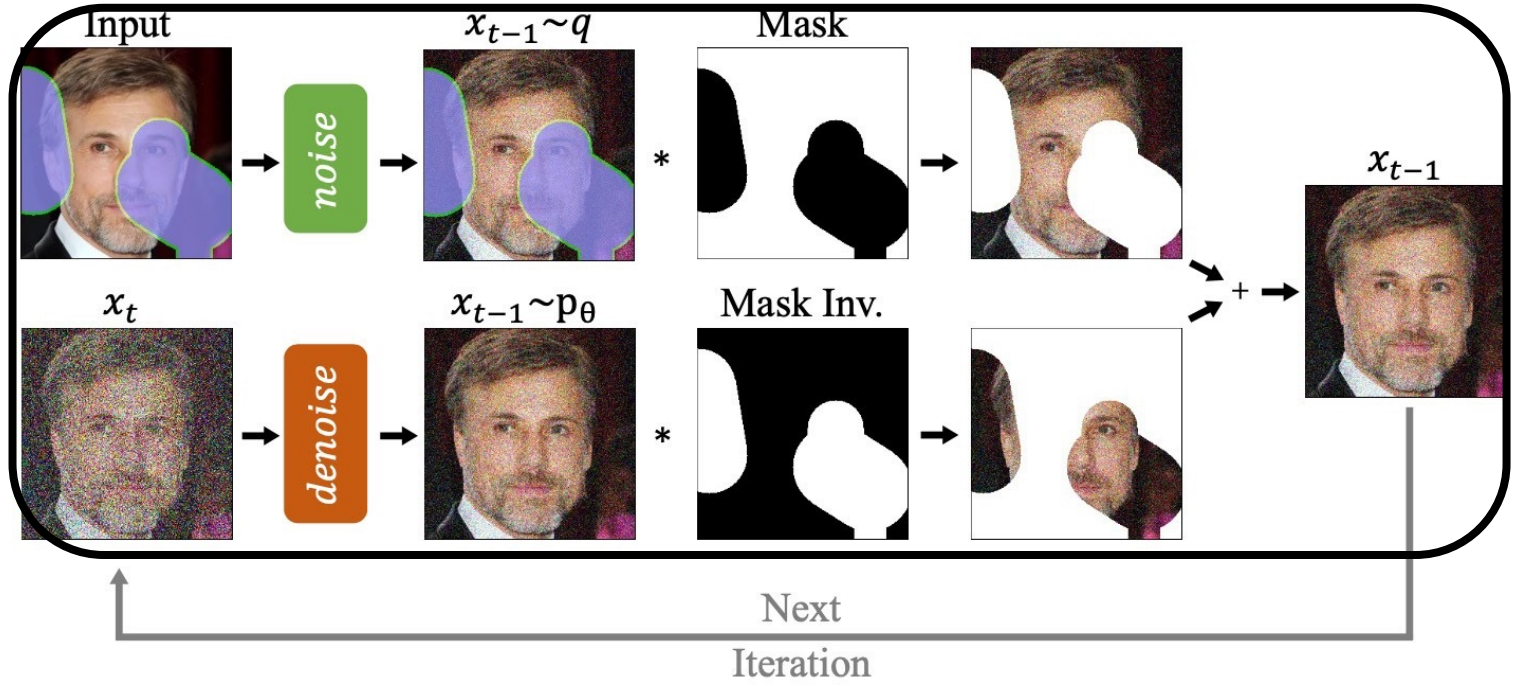
Background

從最乾淨的 x_0 出發

想生成的部分

Foreground

從最 noisy 的 x_T 出發



RePaint – Resampling

反覆 refine 邊界一致性的技巧，在每個 denoising time step t，不是只 denoise 一次，而是重複 U 次“前進-後退”來改善 inpainting 效果

Algorithm 1 Inpainting using our RePaint approach.

1: $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: **for** $u = 1, \dots, U$ **do**
4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\epsilon = \mathbf{0}$
5: $x_{t-1}^{\text{known}} = \sqrt{\bar{\alpha}_t} x_0 + (1 - \bar{\alpha}_t)\epsilon$
6: $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
7: $x_{t-1}^{\text{unknown}} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(x_t, t) \right) + \sigma_t z$
8: $x_{t-1} = m \odot x_{t-1}^{\text{known}} + (1 - m) \odot x_{t-1}^{\text{unknown}}$
9: **if** $u < U$ and $t > 1$ **then**
10: $x_t \sim \mathcal{N}(\sqrt{1 - \beta_{t-1}} x_{t-1}, \beta_{t-1}\mathbf{I})$
11: **end if**
12: **end for**
13: **end for**
14: **return** x_0

U 次 resampling

已知區域加 noise 到 $t - 1$

未知區域 denoise 一步 $(t \to t - 1)$

合併已知和未知區域

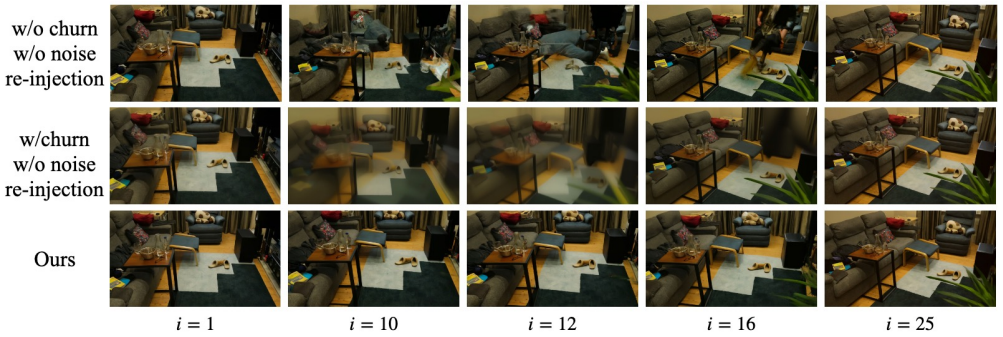
如果不是最後一個 iteration，重新加 noise 回 t

RePaint – Resampling

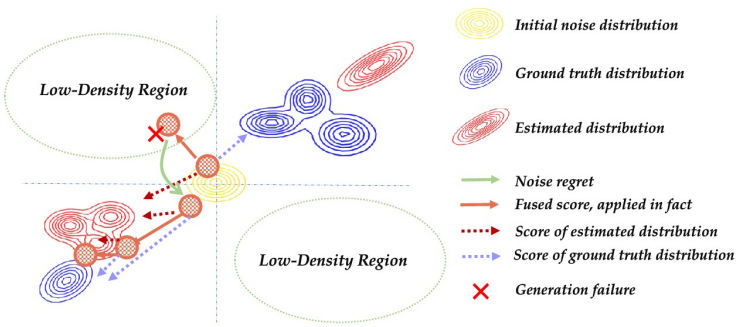
一般的 DDIM 反向去噪就 50 步，但步數這麼少不足以讓邊界協調
那就在同一個 denoising time step 反覆多做幾步

Resampling/Noise Re-Injection/Noise Regret

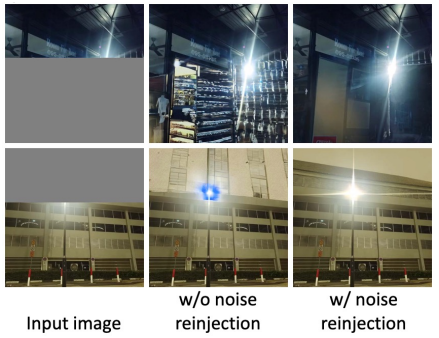
同樣的操作很常見，但有不同的名字



Time-Reversal [ECCV'24]
Noise re-injection



Be-Your-Outpainter [ECCV'24]
Noise regret



LightsOut [ICCV'25]
Noise reinjection

RePaint

不需任何訓練
就可以把一個原本目的是生成的 diffusion model
轉變任務為 inpainting

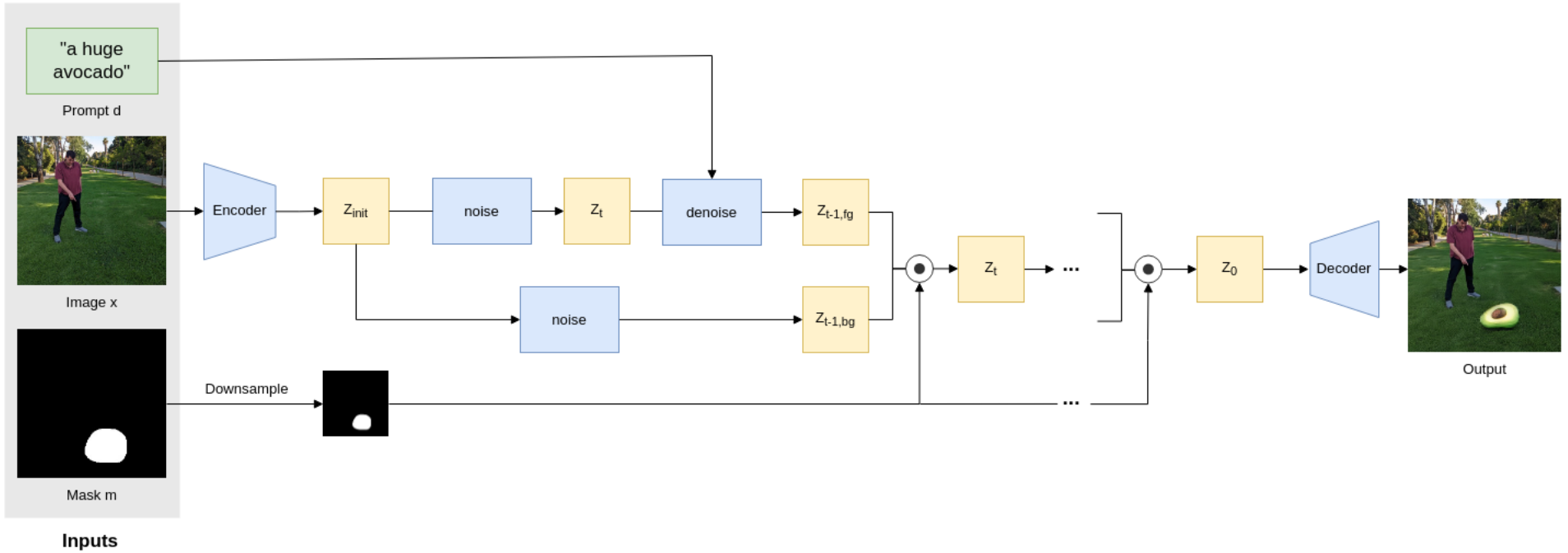
Results with different masks

Outpainting

Blended Latent Diffusion

Avrahami, Omri, Ohad Fried, and Dani Lischinski. “Blended Latent Diffusion.” *ACM TOG*. 2023.

Blended Latent Diffusion



基本上就是 latent 版本的 RePaint

Avrahami, Omri, Ohad Fried, and Dani Lischinski. “Blended Latent Diffusion.” *ACM TOG*. 2023.

Assignment 2

Diffusers

D🧨ffusers

🤗 Diffusers is the go-to library for state-of-the-art pretrained diffusion models for generating images, audio, and even 3D structures of molecules. Whether you're looking for a simple inference solution or training your own diffusion models, 🤗 Diffusers is a modular toolbox that supports both. Our library is designed with a focus on usability over performance, simple over easy, and customizability over abstractions.

🤗 Diffusers offers three core components:

- State-of-the-art diffusion pipelines that can be run in inference with just a few lines of code.
- Interchangeable noise schedulers for different diffusion speeds and output quality.
- Pretrained models that can be used as building blocks, and combined with schedulers, for creating your own end-to-end diffusion systems.